

Possible topics of final projects for COGS 560

1. Pick a reasoning task and apply inference with different LLMs and perform an analysis or comparison across model families and sizes.
2. Fine-tuning an open-weight decoder model (LLama, Gemma, Qwen, etc.) using LoRA for a chosen reasoning task and dataset. (Optional)
3. Writing a term paper about a reasoning task or phenomena related to topics discussed in class. You can adopt either the concept of a positional paper or a survey paper on previous studies on a given problem. For example,
 - Are LLMs the road to AGI or just a stochastic parrot?
 - Can reasoning be achieved with reinforcement learning methods?
 - What is visual reasoning? A collection of visual reasoning methods or benchmarks

You can structure your term paper in the ACL template and submit it to the review cycle and have a chance to obtain official feedback. The next submission deadline is May 25th, 2026! (<https://aclrollingreview.org/dates>)

4. Propose your own project (please discuss with us first).

The term project needs to be at least 8 pages and include a detailed description of the problem, reasoning benchmark, included models in analysis, experiments, and results. Include any figures, references, formulas, and additional details describing the task and your developed system. You can also add theoretical and practical details of your findings. Please also submit any code, notebooks and related material for final grading. You can perform the projects as individuals or groups of two. In case of multiple students you need to inform explicitly the individual responsibilities of each student.

Please inform your decision by April 25th, 2026.

The deadline for submitting your final reports is June 15th, 2026. Failure to submit by this date will result in NI or F grade.